Generative AI in Protein Engineering using Large Language Diffusion Models

A thesis submitted to the
University of Crete & FORTH
for the award of the degree

of

Master of Science

by

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Dedicated to,

my beloved grandparents, whose love and guidance continue to inspire me, despite their absence.

Abstract

This thesis investigates how generative AI can be used to design new protein sequences, focusing on synthetic antimicrobial peptides (AMPs) and ancestral protein reconstruction. It explores three main approaches: interpolating between known proteins using their embeddings, generating new sequences from scratch using diffusion models, and reconstructing ancestral proteins based on evolutionary models.

To guide and evaluate the generated sequences, we use models like AlphaFold from DeepMind for structural prediction and AMP-specific classifiers for functional assessment. Overall, the study highlights how modern generative models, especially diffusion-based ones, can be applied in protein engineering to create meaningful and potentially useful biological sequences.

Recent advances in deep generative models have significantly enhanced the potential for in silico protein design. A powerful pretrained diffusion-based model is EvoDiff [11], developed by Microsoft Research, which combines evolutionary-scale sequence data with the flexible conditioning capabilities of diffusion models to generate diverse and biologically meaningful proteins directly in sequence space.

Keywords: Protein Engineering, Large Language Models, Diffusion Models, Deep Generative AI, Evolutionary Scale Modeling, Ancestral Protein Reconstruction.

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List of Abbreviations

NLP: ***Natural Language Processing.***

LLM: ***Large Language Model.***

PLM: ***Protein Language Model.***

MSA: ***Multiple Sequence Alignment.***

aa: ***Amino Acid.***

ASR: ***Ancestral Sequence Reconstruction.***

ESM: ***Evolutionary Scale Modeling.***

AMPs: ***Antimicrobial Peptides.***

LoRA: ***Low Rank Adaptation.***

pTM: ***Predicted Template Modeling Score.***

pLDDT: ***Predicted Local Distance Difference Test.***

pIDDT: ***Predicted Inter-residue Distance Difference Test.***

KPIs: ***Key Performance Indicators.***

1 Introduction

1.1 Generative AI in Protein Engineering

Since the 2018 Nobel Prize recognised directed evolution as a breakthrough method for tailoring enzymes [5], machine-learning models have accelerated the same idea *in silico*[1]. Autoregressive language models, graph networks, and, most recently, diffusion frameworks such as the powerful pretrained model **EvoDiff** [11] learn from evolutionary-scale data to propose sequences with user-defined functions in minutes rather than months. Commercial groups are already deploying these tools: Arzeda employs biophysics-informed generative models to craft laundry enzymes and sweetener proteins for partners like Unilever [81], while Ginkgo Bioworks has released an API that exposes foundation protein models to external developers [14]. In drug discovery, Insilico Medicine's AI-designed small molecule ISM001-055 reached Phase IIa trials for idiopathic pulmonary fibrosis in just thirty months from target selection, demonstrating the clinical pace achievable with generative design [20].

Beyond therapeutics, generative AI is tackling sustainability challenges. Plastic-degrading enzymes evolved by Carbios now depolymerise PET at industrial scale, enabling the world's first PET biorecycling plant [7]. A key biocatalyst in this effort is the PET hydrolase *PETH_PISS1* (UniProt A0A0K8P6T7), whose crystal structure, expressed in *Escherichia coli*, revealed active-site residues ripe for ML-guided optimisation [30]. Iterative design cycles have already produced variants that outperform natural PETases in both activity and thermal stability [39]. Similar workflows search large metagenomic datasets for new scaffolds and then refine them with deep learning; Protera, for example, uses a deep-learning platform to discover food-grade preservative proteins in partnership with ICL [6]. Underpinning many of these campaigns is the versatility of *E. coli* as an expression chassis, where recent high-throughput automation protocols further compress build–test cycles [41].

Collectively, these advances show how data-driven protein engineering is already reshaping chemical manufacturing, environmental remediation, and medicine, and they point to an even broader design space that diffusion and foundation models are beginning to unlock.

[1] *In silico* means performing experiments or studies inside a computer through digital simulation rather than in a physical lab.

1.2 Related Work

Early sequence–only language models such as ProGen and its successors showed that large-scale autoregressive transformers can generate proteins with measurable activity across diverse families [49]. More recent foundation models, including ESM-3 [24], which simulates hundreds of millions of years of evolution in silico, and backbone-conditioned design tools like ProteinMPNN, demonstrate that pre-trained networks can capture remote homology and rapidly propose functional variants [8, 80]. Despite their success, these approaches rely on a single forward pass and therefore struggle to provide fine-grained control over specific structural or phylogenetic constraints.

Diffusion-based frameworks address this limitation by iteratively refining noise into valid proteins. RFdiffusion generates high-fidelity backbones in 3-D coordinate space and has yielded experimentally validated binders and enzymes [76]. Complementary sequence-space models broaden design diversity: EvoDiff, central to this thesis, leverages evolutionary-scale datasets to create structurally plausible sequences with tunable conditioning [11]. Extensions such as Chroma [57] and ProteinGenerator [15] jointly generate sequence and structure, while TaxDiff incorporates taxonomic labels for species-specific control [46]. New specialised variants, including LigandMPNN for small-molecule contexts [36], illustrate a rapid trend toward domain-aware, controllable diffusion models that balance structural accuracy with evolutionary breadth.

1.3 Scope

The study can be divided into three main sections.

The first part focuses on protein interpolation, where we encode two known proteins with specific characteristics, connect them mathematically through their embeddings, and then decode the result to produce new synthetic proteins. These newly generated sequences are expected to inherit general properties from the original proteins, aiming for a more generalized design.

In the second part, we utilize EvoDiff, a family of diffusion models developed by Microsoft, to generate proteins unconditionally from pure noise. These sequences are then ranked using three main approaches. The first is an open-source tool called CAMP [71], which uses SVM and Random Forest classifiers to predict whether a sequence is antimicrobial (AMP) or not. The second approach is based on a model fine-tuned using LoRA optimization, and the third relies on analyzing the physicochemical structure of the proteins to estimate the likelihood of them being AMPs.

The third part deals with the ancestral reconstruction of given proteins. For this, we employ the FIRE-PROT model, which uses evolutionary substitution models (such as JTT[2], WAG[3], and LG[4] [29, 78, 38].) combined with maximum likelihood estimation to infer the most probable amino acid at each ancestral node of a phylogenetic tree. Once the tree is constructed, we trace back through evolutionary time, mask amino acids with low probability, and use these partially masked sequences as conditional inputs to EvoDiff's diffusion model. This enables us to generate new proteins that preserve ancestral information while potentially introducing novel and functional variation.

All code, scripts, and generated data are available on my GitHub

1.4 Methodological Tools

For this thesis we rely heavily on state-of-the-art pretrained models that have reshaped modern protein engineering. Chief among them are Meta AI's *Evolutionary Scale Modeling* (ESM) language models and DeepMind's *AlphaFold* structure predictor, which together provide sequence-level and structure-level priors for downstream design.

1.4.1 ESM

Evolutionary Scale Modeling (ESM) refers to a suite of Transformer-based protein–language models released by Meta AI's FAIR team [66, 45]. Trained on up to 2.5 billion non-redundant sequences, these models treat amino-acid strings as sentences and learn contextual relationships via the masked-language objective introduced in BERT. Formally, for an input sequence $\mathbf{x} = (x_1, \ldots, x_L)$, a random subset $\mathcal{M}$ of positions is replaced by a mask token $\langle\text{mask}\rangle$. The model then minimises the cross-entropy loss

$$\mathcal{L}_{\text{MLM}} = -\sum_{i\in\mathcal{M}} \log p_\theta\big(x_i \,\big|\, \mathbf{x}_{\setminus i}\big),$$

where θ are the network parameters and $\mathbf{x}_{\setminus i}$ denotes the sequence with position i masked out. Self-attention weights implicitly encode residue–residue couplings; higher-layer heads correlate strongly with contact maps and functional motifs [62]. Because the model is trained without structural labels, it scales effortlessly with new metagenomic data and generalises to remote homologues unseen during training.

ESM embeddings power a diverse set of downstream tasks: (i) unsupervised contact prediction that

[2]Jones–Taylor–Thornton model

[3]Whelan and Goldman model

[4]Le and Gascuel model

rivals co-evolutionary methods; (ii) zero-shot assessment of variant effects; (iii) rapid remote-homology search that outperforms BLAST on speed and sensitivity [23]; and (iv) end-to-end structure prediction—ESMFold—achieving near-AlphaFold accuracy in milliseconds [45]. These capabilities make ESM a universal sequence engine for protein interpolation, clustering, and functional annotation pipelines.

1.4.2 AlphaFold

AlphaFold revolutionised structural biology by predicting atomic coordinates from sequence with near-experimental accuracy [34]. Its architecture combines evolutionary information and geometric reasoning in two stages: the *Evoformer* and the *Structure Module*.

Multiple-sequence alignments (MSAs) $\mathbf{M} \in \mathbb{R}^{N_{\text{seq}} \times L \times d}$ and template pair features $\mathbf{P} \in \mathbb{R}^{L \times L \times d'}$ are iteratively updated through axial self-attention, triangular multiplicative updates, and outer-product operations. Let $\mathbf{z}_t$ be the pair representation at iteration t; a single triangular update can be written

$$\mathbf{z}_{t+1} = \mathbf{z}_t + \text{Tri}\big(\mathbf{z}_t\big),$$

where $\text{Tri}(\cdot)$ denotes a combination of triangle-attention and triangle-multiplication layers that propagate geometric constraints across residue pairs [2]. After T recycling iterations, the refined pair features serve as a prior over inter-residue distances and orientations.

Given pairwise distance logits $\mathbf{d}$ and initial backbone frames, the structure module predicts 3-D coordinates $\mathbf{X} \in \mathbb{R}^{L \times 3}$ by minimising a frame-aligned point error (FAPE) and torsion angle losses. Confidence is reported via predicted Local Distance Difference Test (pLDDT) and predicted TM-score (pTM), enabling automated quality assessment.

AlphaFold released an open database (AF-DB) with > 250 million predicted structures [16], transforming fields from drug discovery to enzyme engineering. Researchers now routinely feed de novo sequences into AlphaFold to triage folding stability, locate functional pockets, and refine design hypotheses. The upcoming AlphaFold 3 extends the framework to joint protein–ligand and nucleic-acid complexes using a diffusion backbone [21], underscoring the model's central role in next-generation bio-design.

1.4.3 Diffusion Models

Diffusion models form a broad class of generative latent–variable models that transform simple noise into complex data through a learned time-reversal of a stochastic diffusion process. Introduced for images as *Denoising Diffusion Probabilistic Models* (DDPMs) [26] and later generalised to continuous-time score-based frameworks [68], they now underpin state-of-the-art synthesis in domains ranging from vision to protein design.

Forward (diffusion) process. Given a data distribution $q(\mathbf{x}_0)$, the forward process gradually perturbs data with Gaussian noise over T discrete time-steps:

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\big(\sqrt{\alpha_t}\,\mathbf{x}_{t-1},\, (1-\alpha_t)\mathbf{I}\big), \quad 1 \le t \le T,$$

where $(\alpha_t)_{t=1}^{T}$ is a variance schedule with $\alpha_t \in (0,1)$. Marginalising, one obtains a closed form

$$q(\mathbf{x}_t \mid \mathbf{x}_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big), \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s.$$

Reverse (denoising) process. Sampling requires learning a parametric estimate $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathcal{N}\big(\boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \sigma_t^2\mathbf{I}\big)$, where $\boldsymbol{\mu}_\theta$ is predicted by a neural network—commonly via a noise-prediction head $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$. Training minimises a weighted variational bound; a simplified yet empirically effective loss is

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}}\big\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\big(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon},\, t\big)\big\|^2,$$

with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. In the continuous-time limit, the process is described by a stochastic differential equation (SDE) $\mathrm{d}\mathbf{x}_t = f(t)\,\mathbf{x}_t\,\mathrm{d}t + g(t)\,\mathrm{d}\mathbf{w}_t$, whose time-reversed dynamics can be numerically integrated once the *score function* $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t)$ is learned [68].

Diffusion models exhibit: (i) *Likelihood tractability*, enabling principled evaluation; (ii) *Stable training* via simple ℓ_2 objectives; (iii) *Flexible conditioning*, e.g. classifier or motif guidance, which is crucial for controlled protein design.

Structure-space models such as RFdiffusion generate atomic backbones with near-experimental accuracy [77]. Sequence-space variants, including EvoDiff, central to this thesis, apply the same principles directly to amino-acid chains, using categorical or continuous relaxations of sequence tokens. Hybrid frameworks (e.g. Chroma [56]) jointly diffuse sequence and structure, demonstrating the versatility of the approach. The success of these models lies in their ability to explore vast design spaces while incorporating biologically motivated constraints during the denoising trajectory.

2 KPIs for Antimicrobial Peptides

Proteins can be classified as antimicrobial peptides (AMPs) through a variety of computational methodologies that exploit sequence statistics, machine-learning classifiers, and physicochemical heuristics. In this work we focus on three complementary approaches: a supervised classifier from the CAMP database, a low-rank–adapted ESM model fine-tuned for AMP recognition, and a physicochemical scoring scheme derived from well-established empirical rules.

2.1 CAMP Classifier

The *Collection of Anti-Microbial Peptides* (CAMP) database houses $>$ 10,000 curated AMP sequences and associated metadata [71, 37]. CAMP exposes a webserver that applies Support Vector Machines (SVM) and Random Forests, to predict AMP activity. For an input peptide represented by a feature vector $\boldsymbol{\phi}(\mathbf{x})$ (AAC, dipeptide composition, charge, hydrophobicity, etc.), an SVM learns a separating hyperplane

$$f(\mathbf{x}) = \operatorname{sign}\big(\mathbf{w}^{\top}\boldsymbol{\phi}(\mathbf{x}) + b\big),$$

where $\mathbf{w}$ and b are obtained by minimising the hinge loss with a regularisation term $\frac{1}{2}\|\mathbf{w}\|^2$.

Random Forest augments this by training an ensemble $\{h_t\}_{t=1}^{T}$ of decision trees on bootstrap samples and returning the majority vote $\operatorname{mode}\{h_t(\mathbf{x})\}$ [13]. Benchmark studies report accuracies of 90–94% on independent AMP datasets, making CAMP a widely adopted baseline for *in silico* screening [73].

2.2 Low Rank Adaptation of ESM-2 for AMP Classification

Parameter-efficient fine-tuning with *Low-Rank Adaptation* (LoRA) [27] allows large protein language models to specialise for AMP detection without updating all weights. Given a pretrained ESM-2 model with parameters θ, LoRA injects rank-$r \ll d$ adapters $\Delta\theta = \mathbf{A}\mathbf{B}^{\top}$ into attention and MLP layers while freezing θ. During training, only $\mathbf{A}, \mathbf{B}$ (with $\mathcal{O}(rd)$ parameters) are optimised via cross-entropy on an AMP/non-AMP token label $y \in \{0, 1\}$:

$$\mathcal{L}_{\text{CE}} = -\big[y \log p_{\phi}(\mathbf{x}) + (1-y)\log(1-p_{\phi}(\mathbf{x}))\big],$$

where $\phi = \theta \cup \{\mathbf{A}, \mathbf{B}\}$. Recent work shows that LoRA-tuned ESM models achieve F1 scores comparable to full fine-tunes while reducing training memory by $\approx$90 % [40, 47]. Such efficiency enables rapid iterative retraining as new AMP labels emerge.

2.3 Physicochemical Scoring

Sequence-derived physicochemical indices provide an interpretable, alignment-free proxy for antimicrobial potency [60]. Key features include:

- **Net charge** at pH 7: $Q = \sum_i n_i q_i$, where n_i is the count of residue i and $q_i \in \{-1, 0, +1\}$ is its side-chain charge. Active AMPs typically satisfy $Q \geq +2$.
- **Hydrophobic ratio** $H = \frac{\#\text{ hydrophobic residues}}{L}$, which modulates membrane insertion.
- **Boman index** $B = \sum_i n_i \Delta G_i / L$, estimating binding free energy to proteins; values between -2 and 3 kcal mol^{-1} often correlate with broad-spectrum activity.

Several studies combine these metrics into a logistic model

$$\Pr(\text{AMP} \mid \mathbf{x}) = \sigma\big(\beta_0 + \beta_1 Q + \beta_2 H + \beta_3 B\big),$$

yielding AUROC ≥ 0.90 on balanced datasets [51, 28]. Though less flexible than deep models, physicochemical scoring offers rapid first-pass filtering and mechanistic interpretability, complementing the machine-learning approaches above.

3 Protein Interpolation

3.1 Protein Embedding with ESM-2

We are using ESM2, a 48-layer Transformer model that maps a protein sequence $x = (x_1, \ldots, x_L)$ to a latent vector $z \in \mathbb{R}^d$. To obtain this representation, we take the average of the final hidden states from the last layer:

$$z = \frac{1}{L} \sum_{i=1}^{L} h_i^{(48)}$$

Encoder. The encoder is the frozen forward pass $f_\theta(x)$, using the standard 21-letter amino acid alphabet. Padding is done to multiples of 128 for batching.

Decoder. Because ESM-2 is not generative, we trained a custom decoder g_ϕ following DiMa [52]. It is a two-layer MLP mapping $z \to \mathbb{R}^{L \times 21}$. The predicted token is selected via $\arg\max$. We trained it on 1.2M UniRef50 sequences.

Reconstruction Check. To assess the accuracy of the decoder, we used the *Levenshtein distance*, a common metric for comparing the similarity between two strings. It works by counting the minimum number of single-character edits, insertions, deletions, or substitutions, needed to transform one sequence into another. This provides a straightforward way to measure how closely a reconstructed sequence matches the original.

For example, consider the original sequence `"MKTPL"` and a reconstructed version `"MKAPL"`. To transform the original into the reconstructed version, only one substitution is needed, replacing 'T' with 'A'. Therefore, the Levenshtein distance between these two sequences is 1.

We applied this metric to 1,000 randomly sampled sequences, comparing each original to its reconstructed counterpart. An average Levenshtein distance of 1.002 across these sequences means that, on average, each reconstructed sequence differs from the original by about one single-character edit, which could be a substitution, insertion, or deletion. This small difference demonstrates the custom ESM decoder's effectiveness in preserving sequence accuracy during the encoding and decoding process.

The figure below illustrates how our procedure works.

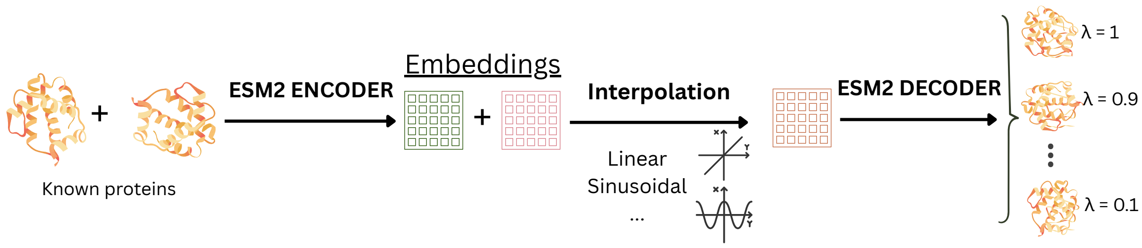


Figure 1: Protein interpolation procedure using ESM2

3.2 Interpolated Synthetic Sequences

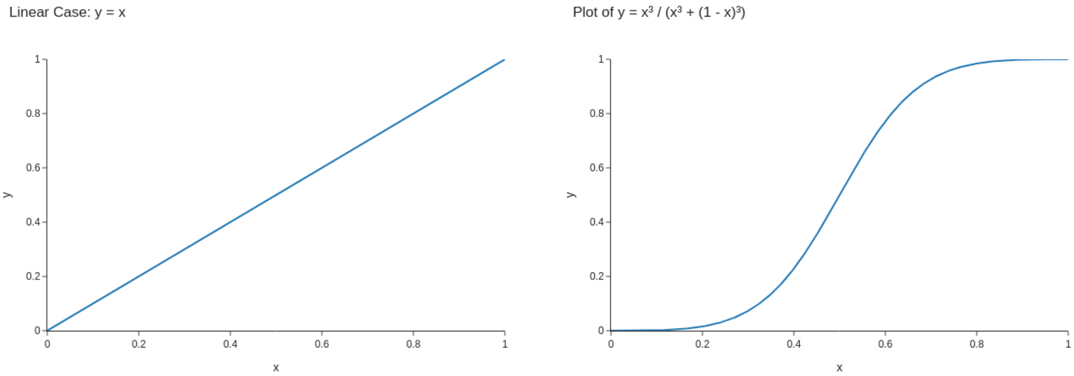
Given two sequences:

- `DRAMP04678`: DSECLKEYGGDVGFGFCAPRIYPSFCVQRCRADKGALSGKCIWGQGSNVKCLCNFCRHEP
- `ADAM_2065`: GIWSSIKNLASKAWNSDIGQSLRNKAAGAINKFVADKIGVTPSQAASMTLDEIVDAMYYD

We compute embeddings z_1 (`DRAMP04678`), z_2 (`ADAM_2065`), then interpolate via:

◇ ***Linear interpolation:*** $z_\lambda^{\text{lin}} = (1-\lambda)z_1 + \lambda z_2$

◇ ***Power-law interpolation:*** $w_\lambda = \dfrac{\lambda^a}{\lambda^a + (1-\lambda)^a}, \quad z_\lambda^{\text{power}} = (1-w_\lambda)z_1 + w_\lambda z_2, \quad \text{with } a = 3$



(a) Linear interpolation function

(b) Power-law interpolation function ($a = 3$)

Figure 2: Interpolation weighting functions for embeddings.

The embeddings, each of shape 60×1280. In the following, we observe the interpolated sequences.

λ	Similarity ↑ (%)	Similarity with target source sequences	CAMP prob.	Decoded sequence
0.00	–	3.33 | 100.00	0.85	DSECLKEYGGDVGFGFCAPRIYPSFCVQRCRADKGALSGKCIWGQGSNVKCLCNFCRHEP
0.10–0.30	98.33	3.33 | 98.33	0.72	MSECLKEYGGDVGFGFCAPRIYPSFCVQRCRADKGALSGKCIWGQGSNVKCLCNFCRHEP
0.40	90.00	13.33 | 88.33	0.77	MSECLKEYGGDKGFGSCAGRILRSFCVQRCRADKGALSGKCISGQGSNVKCLCNFCRHEP
0.45	91.67	21.67 | 80.00	0.74	MSECLKENGADKAFGSCAGRILRSFCVQRCRKDKGALSGKCISGAGSNVKCLCNFCRHEP
0.50	75.00	46.67 | 55.00	0.77	MSWSSIKNLASKAFGSCAGRSLRSKCAQACRKDKGAKSGKCISGAGSNVKCLCVDCRHED
0.55	81.67	65.00 | 36.67	0.84	MIWSSIKNLASKAWNSCIGRSLRNKCAGACRKFVGAKSGKCISQAGSNTKCLCVDCRYED
0.60	73.33	90.00 | 10.00	0.48	MIWSSIKNLASKAWNSSIGQSLRNKAAGAINKFVADKIGVCPSQAASNTLDEIVDARYED
0.70–1.00	90.00	100.00 | 3.33	0.83	GIWSSIKNLASKAWNSDIGQSLRNKAAGAINKFVADKIGVTPSQAASMTLDEIVDAMYYD

Table 1: Linear interpolation: decoded proteins and similarity ↑ to the previous sequence. The third column reports sequence alignment with the initial sequences `ADAM_2065` (left) and `DRAMP04678` (right).

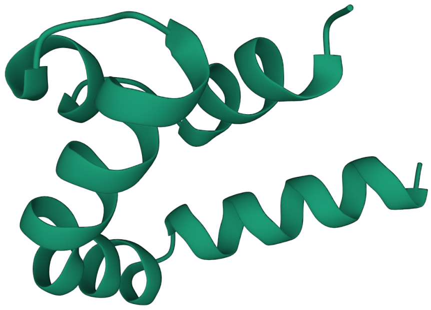
λ	Similarity ↑ (%)	Similarity with target source sequences	CAMP prob.	Decoded sequence
0.00–0.30	–	3.33 | 100.00	0.85	DSECLKEYGGDVGFGFCAPRIYPSFCVQRCRADKGALSGKCIWGQGSNVKCLCNFCRHEP
0.40	98.33	3.33 | 98.33	0.72	MSECLKEYGGDVGFGFCAPRIYPSFCVQRCRADKGALSGKCIWGQGSNVKCLCNFCRHEP
0.45	95.00	8.33 | 93.33	0.73	MSECLKEYGGDVGFGFCAGRILPSFCVQRCRADKGALSGKCISGQGSNVKCLCNFCRHEP
0.50	61.67	46.67 | 55.00	0.77	MSWSSIKNLASKAFGSCAGRSLRSKCAQACRKDKGAKSGKCISGAGSNVKCLCVDCRHED
0.55	48.33	98.33 | 3.33	0.28	MIWSSIKNLASKAWNSDIGQSLRNKAAGAINKFVADKIGVTPSQAASMTLDEIVDAMYYD
0.60–1.00	98.33	100.00 | 3.33	0.83	GIWSSIKNLASKAWNSDIGQSLRNKAAGAINKFVADKIGVTPSQAASMTLDEIVDAMYYD

Table 2: Power-law ($a = 3$) interpolation: decoded proteins and similarity ↑ to the previous sequence. The third column reports sequence alignment with the initial sequences `ADAM_2065` (left) and `DRAMP04678` (right).

The colored rows *highlight* the sequences found in both tables. Furthermore, Random Forest CAMP classifier (the highest-performing CAMP model) was used to evaluate the synthesized proteins. As expected, the two target source proteins received the highest AMP probabilities, confirming their known antimicrobial nature. Most of the synthesized proteins were also classified as AMPs with relatively high probability, except for the case at $\lambda = 0.55$, which was predicted to be a non-AMP by both interpolation methods.

3.2.1 Structural Observations with ESMFold

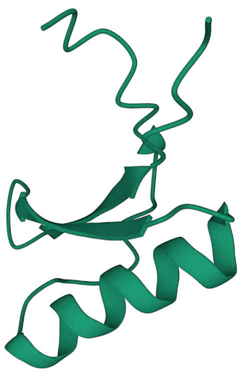
ESMFold was used to predict the 3D structures of both original sequences and the interpolated.



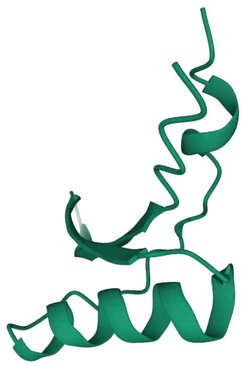
(a) ADAM_2065 with mean pLDDT: 73.5, pTM: 0.45

(b) DRAMP04678 with mean pLDDT: 69, pTM: 0.6

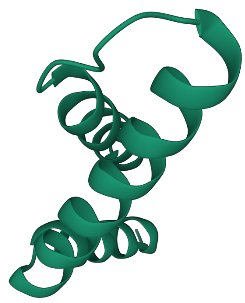
(c) $\lambda = \mathbf{0.45}$ with mean pLDDT: 69.2, pTM: 0.6 `ADAM_2065 similarity:` **8.5%** `DRAM04678 similarity:` **93.5%** Interpolation: Power-Law.



(d) $\lambda = \mathbf{0.5}$ with mean pLDDT: 56, pTM: 0.45 `ADAM_2065 similarity:` **47%** `DRAM04678 similarity:` **55%** Interpolation: Both



(e) $\lambda = \mathbf{0.55}$ with mean pLDDT: 57.5, pTM: 0.45 `ADAM_2065 similarity:` **65%** `DRAM04678 similarity:` **37%** Interpolation: Linear



(f) $\lambda = \mathbf{0.6}$ with mean pLDDT: 68.5, pTM: 0.43. `ADAM_2065 similarity:` **90%** `DRAM04678 similarity:` **10%** Interpolation: Linear

Figure 3: Predicted structures using ESMFold [44]. Input proteins (top); interpolated proteins (bottom).

The Figure 3 above illustrates a clear structural progression from the β-sheet–dominated `DRAMP04678` to the largely α-helical `ADAM_2065`. At $\lambda = 0.45$, the model still resembles the DRAMP fold: three well-packed β-strands buttressed by a short helix, a high confidence profile (mean pLDDT $\approx$ 69; pTM $\approx$ 0.60), and >90% sequence identity to DRAMP indicate a stable DRAMP-like topology. Shifting to $\lambda = 0.50$ produces a true hybrid: the central sheet and helix coexist, but long unstructured loops appear and the confidence drops markedly (pLDDT $\approx$ 56, pTM $\approx$ 0.45), mirroring the balanced 47% / 55% sequence split between the parents.

Beyond this midpoint the structure steadily adopts ADAM features. At $\lambda = 0.55$ the β-sheet fragments, the helix elongates, and the sequence now leans 65% toward ADAM; confidence begins to rebound. By $\lambda = 0.60$ the protein is almost entirely helical, retaining only a vestigial β-turn, and the metrics (pLDDT $\approx$ 69, pTM $\approx$ 0.43) approach those of the ADAM endpoint with 90% sequence identity. Together, these snapshots show that linear interpolation yields a smooth yet asymmetric morph: modest sequence changes near DRAMP trigger large fold rearrangements, whereas once the helix dominates, additional ADAM-like mutations merely extend a pre-formed helical bundle.

4 Protein Generation using EvoDiff

Recent work on sequence–to–function modelling has shown that diffusion models can be transferred from continuous domains (images, audio) to the discrete space of protein sequences while still exploiting deep evolutionary priors. **EvoDiff** [10] extends the classical denoising diffusion probabilistic model (DDPM) framework by treating a protein family's multiple-sequence alignment (MSA) as an *evolutionary time series*. The forward–diffusion process progressively corrupts a sequence toward the mutation spectrum observed in nature, and a neural decoder is trained to reverse this corruption. In effect, EvoDiff learns to "walk backwards" along plausible mutational trajectories, allowing conditional generation of novel, yet evolutionarily consistent, proteins.

4.1 Overview of EvoDiff

For this thesis we focus on the two principal diffusion variants implemented in the public `evodiff` package that Microsoft provides[5]:

1. **Order-Agnostic Autoregressive Diffusion (OADM)**: at each forward step a single residue is

[5] The repository can be found here https://github.com/microsoft/evodiff

replaced by a special `<mask>` token, chosen *uniformly* without regard to sequence position. After L steps all positions are masked. The reverse process iteratively infers the most probable amino acid at the position currently masked, yielding an autoregressive sampler that is permutation invariant over positions.

2. **Discrete Denoising Diffusion Probabilistic Model (D3PM)**: a direct adaptation of DDPMs to categorical data. The forward kernel stochastically mutates residues according to an empirically derived amino-acid transition matrix, while the reverse kernel denoises by predicting a categorical distribution over the 20-letter alphabet. D3PM provides a principled likelihood and typically achieves higher sequence diversity than OADM.

Both models leverage a transformer backbone pre-trained on UniRef50, enabling the sampler to respect long-range epistatic couplings captured from natural evolution.

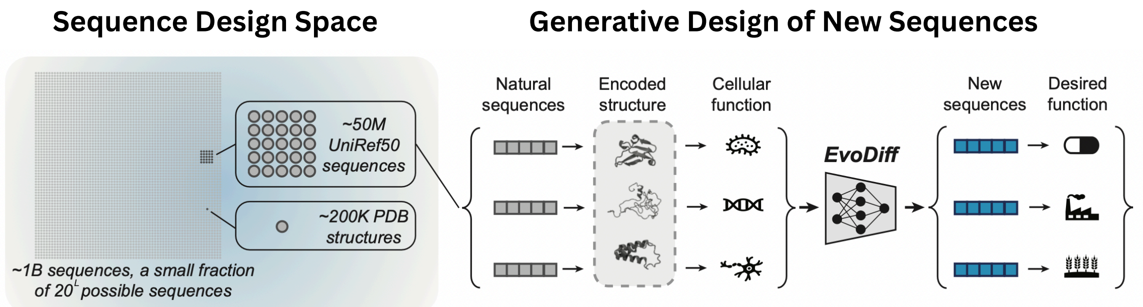


Figure 4: ***Protein sequence design through evolutionary diffusion modeling***. (Left) Evolution has only explored a very small subset of the vast space of possible protein sequences, and structural information is available for even fewer. (Right) EvoDiff is a discrete diffusion-based generative model trained on naturally occurring protein sequences. By sampling from it, one can create novel protein sequences with potentially useful biological properties [10].

4.1.1 Order-Agnostic Autoregressive Diffusion (OADM)

Order-Agnostic Autoregressive Diffusion (OADM) is a discrete diffusion model where, at each forward diffusion step t, a single amino acid token in the sequence x_t is replaced with a special mask token, in a randomly chosen order. The forward process continues until all positions in the sequence are masked. Let L denote the length of the sequence. Then after $T = L$ steps, the full sequence becomes masked:

$$x_0 \xrightarrow{q_1} x_1 \xrightarrow{q_2} \cdots \xrightarrow{q_T} x_T = \texttt{[MASK]}^L$$

where each $q_t(x_t|x_{t-1})$ replaces one unmasked token with `[MASK]`.

The reverse process is modeled by a neural network $p_\theta(x_{t-1}|x_t)$ trained to predict the previously masked token at each step. This process is order-agnostic, meaning the network learns to denoise regard-less of the masking order, enabling flexible conditional generation via inpainting:

$$p_\theta(x_0|x_T) = \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$

4.1.2 Discrete Denoising Diffusion Probabilistic Models (D3PM)

In Discrete Denoising Diffusion Probabilistic Models (D3PM), the forward process adds noise by randomly mutating amino acids in the sequence using a Markov chain defined by a transition matrix Q_t. These transitions model realistic substitution probabilities (e.g., uniform or BLOSUM62-based). Over time, the sequence becomes increasingly corrupted until it resembles a uniform distribution:

$$q(x_t|x_{t-1}) = Q_t(x_t|x_{t-1})$$

After T steps, x_T becomes nearly indistinguishable from a random sample over the amino acid vocabulary $\mathcal{A}$.

The reverse denoising process is parameterized by a neural network to approximate:

$$p_\theta(x_{t-1}|x_t)$$

with the full generative process:

$$p_\theta(x_0|x_T) = \prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$$

Unlike OADM, D3PM permits changes to all tokens at every step, making it suitable for guided generation by iteratively refining all residues.

4.2 Unconditional Generation of Protein Sequences

Unconditional generation refers to producing protein sequences without any external guidance or constraints. In this setting, EvoDiff begins with an entirely corrupted sequence, either fully masked (in OADM) or randomly initialized (in D3PM), and reconstructs a plausible protein sequence by reversing the corruption process. This iterative denoising is learned from large-scale protein sequence datasets, allowing the model to generate novel sequences that resemble natural proteins in terms of structure, function, and diversity. Such generation is valuable for exploring unexplored regions of sequence space and for creating candidate proteins for downstream evaluation or design tasks.

Using the OADM, we employed EvoDiff to unconditionally generate 500 protein sequences for each length from 5 to 50, in increments of 5 (i.e., lengths 5, 10, 15, ..., 50). In addition, we generated 500 sequences for each of the following lengths: 22, 27, 32, 37, 42 and 47. The same procedure was repeated using the D3PM model, resulting in a total of ***16,000*** generated sequences. Both generative models were instantiated with 640 million parameters each.

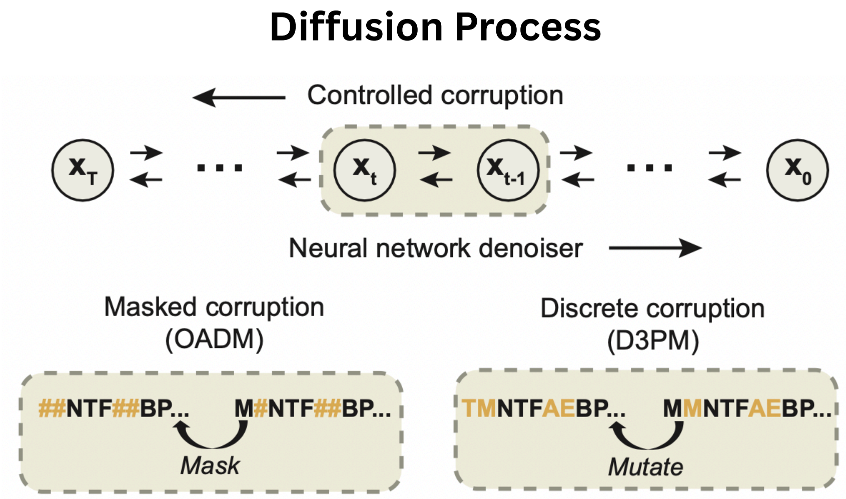


Figure 5: Discrete diffusion models involve a structured noise addition phase and a trained denoising mechanism. In the masked corruption approach (bottom left), sequence tokens are randomly masked without following a specific order. In the discrete corruption strategy (bottom right), noise is introduced through a Markov process governed by a transition matrix that reflects typical amino acid substitution patterns [10].

4.3 Evaluation of Generated Proteins

To judge whether the sequences produced by *EvoDiff* resemble functional antimicrobial peptides, we evaluate them using three complementary approaches.

- ***CAMP classifiers***: Random Forest and Support Vector Machine models trained on curated AMP datasets.
- ***LoRA−ESM***: a fine-tuned lightweight adapter on top of ESM2 (`esm2_t36_3B_UR50D`), producing an AMP score.
- ***Physicochemical scoring***: a rule-based approach based on amino acid composition and biophysical thresholds.

Each method is applied independently to the sequences generated by both OADM and D3PM diffusion models.

4.3.1 AMP Scoring

Below, we visualize the number of predicted AMPs across different sequence lengths.

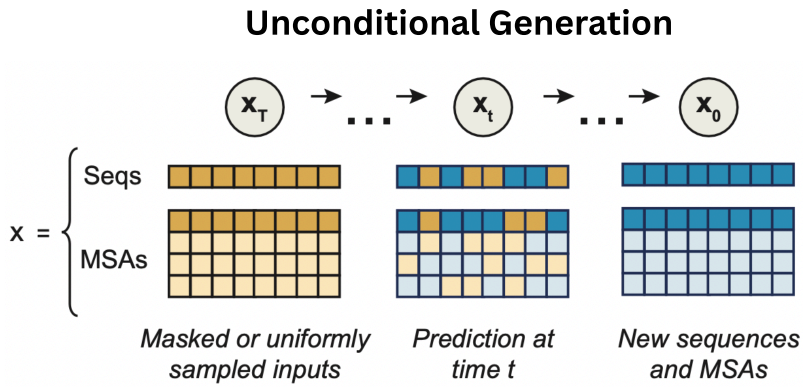


Figure 6: ***EvoDiff supports unconditional generation of protein sequences or MSAs***. Beginning with masked or randomly initialized inputs x_T, the model reconstructs realistic outputs x_0 by progressively reversing the corruption steps through iterative denoising [10].

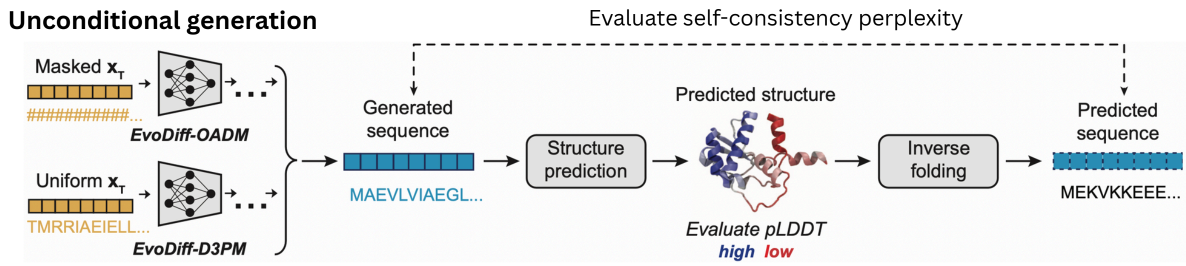
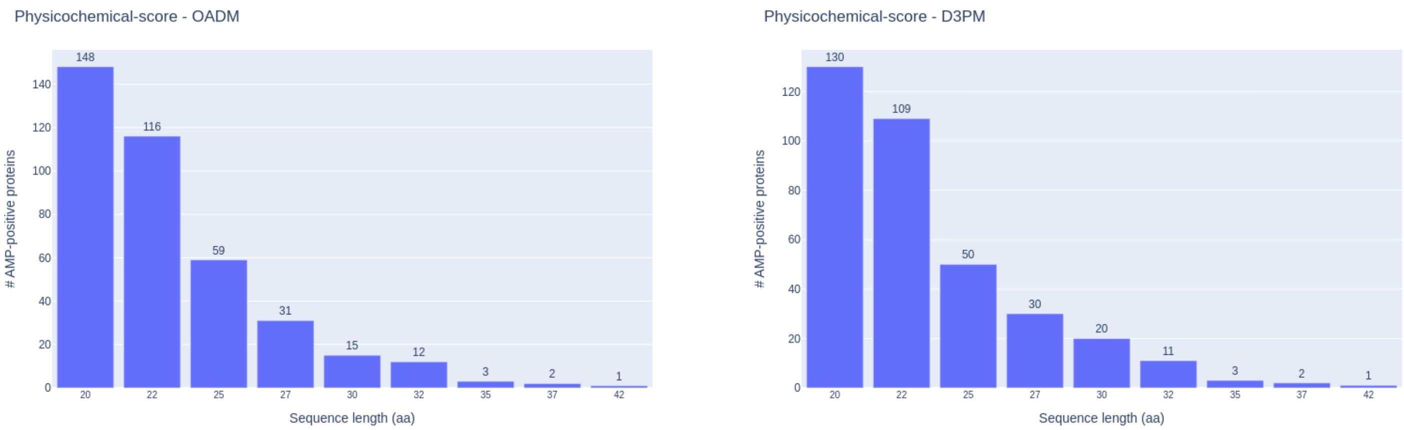


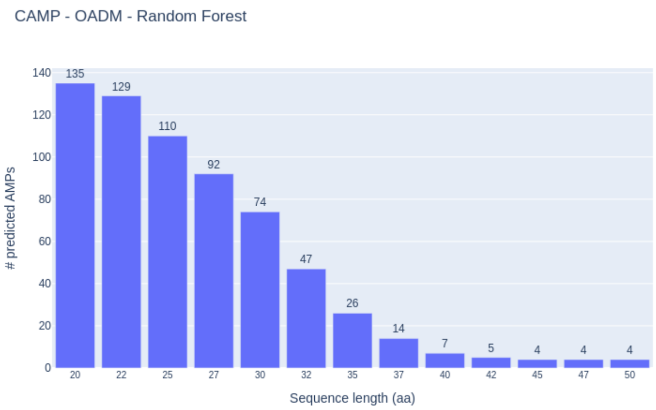
Figure 7: ***EvoDiff generates realistic protein sequences that are both realistic and likely to adopt valid structures***. Shown is the evaluation pipeline used to assess the structural foldability and internal consistency of the sequences generated by EvoDiff [10].



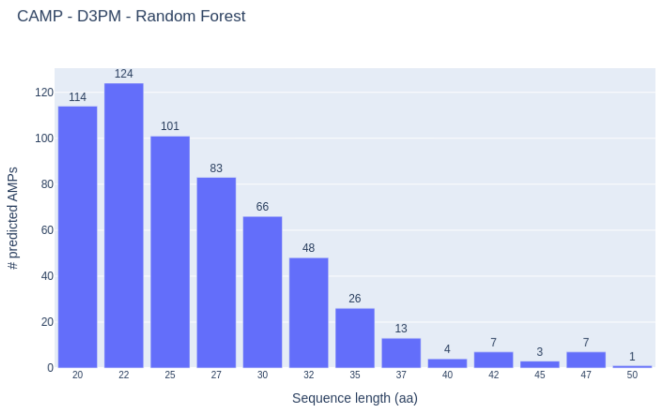
(a) Physicochemical (OADM)

(b) Physicochemical (D3PM)

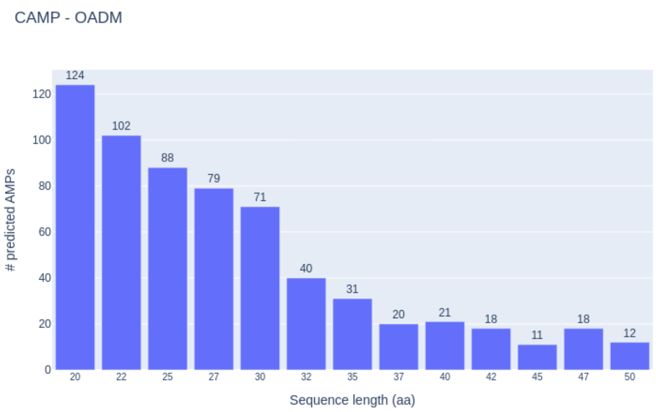
Figure 8: Predicted AMPs using physicochemical filtering.



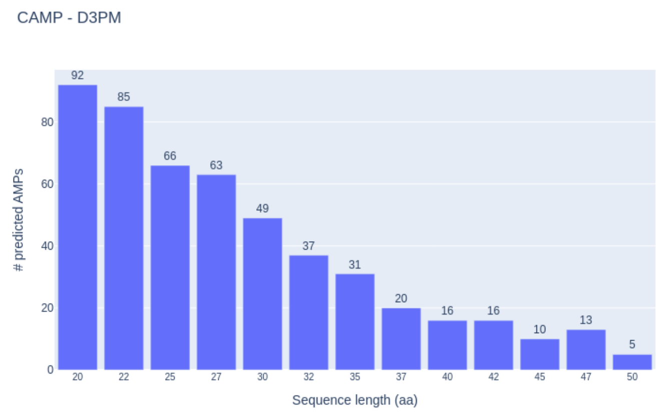
(a) CAMP–RF (OADM)



(b) CAMP–RF (D3PM)

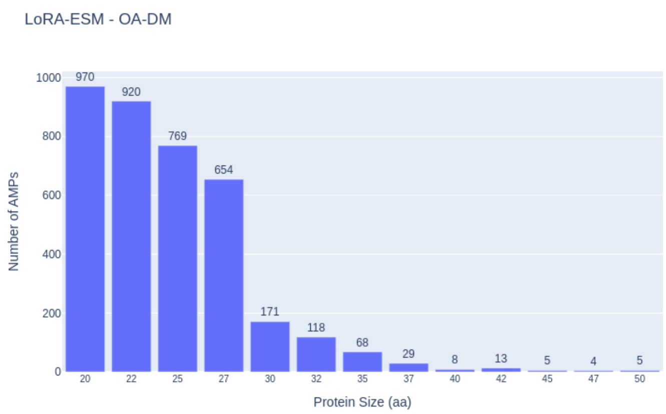


(c) CAMP–SVM (OADM)

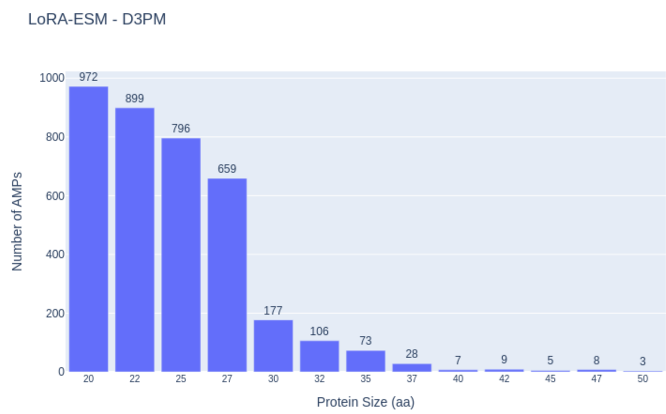


(d) CAMP–SVM (D3PM)

Figure 9: Predicted AMPs using CAMP classifiers.



(a) LoRA–ESM (OADM)



(b) LoRA–ESM (D3PM)

Figure 10: Predicted AMPs using the LoRA–ESM classifier.

4.3.2 AMP Classifiers Comparison

The three AMP scoring methods provide a consistent yet distinct view of how the diffusion models perform. In the LoRA–ESM results (Fig. 10), both OADM and D3PM generate a very high number of predicted AMPs at short lengths, close to 1000 sequences at 20 residues, followed by a steep drop beyond 30–32 aa. This strong ”length dependency” is noticeably more pronounced than in the CAMP or physicochemical results. One likely explanation is that the LoRA adapter was trained on AMP datasets dominated by short peptides. Since LoRA updates only a small number of parameters, it may learn superficial length-associated patterns without adjusting the deeper contextual representation. As a result, it tends to over-score shorter sequences, possibly mistaking length for antimicrobial potential.

By contrast, the CAMP classifiers (Fig. 9) show a more gradual decline across the sequence length range. The RF and SVM models identify around 110–135 AMPs at 20 aa for OADM, decreasing to just a few at 50 aa. The difference between OADM and D3PM is also smaller compared to LoRA, suggesting that CAMP's engineered features reduce length bias. The physicochemical (Fig. 8) is the most selective, reporting much lower AMP counts overall and showing minimal preference between generators.

Taken together, these comparisons suggest that OADM has a slight edge in producing short, charge-rich peptides, but that LoRA's high hit-rate at 20–27 aa is likely inflated by length over-fitting; the CAMP and physicochemical metrics provide a more balanced estimate of genuinely AMP-like candidates across the full size range.

4.3.3 Uniform Manifold Approximation and Projection (UMAP)

UMAP is a non-linear dimensionality-reduction algorithm that builds a graph of high-dimensional relationships and optimises a low-dimensional layout that preserves both local and global structure. Unlike PCA, which assumes linearity, or t-SNE, which distorts global distances, UMAP balances both neighbourhood fidelity and manifold continuity, making it particularly suitable for visualising complex biological embeddings. The figure below shows how the UMAP plots are generated.

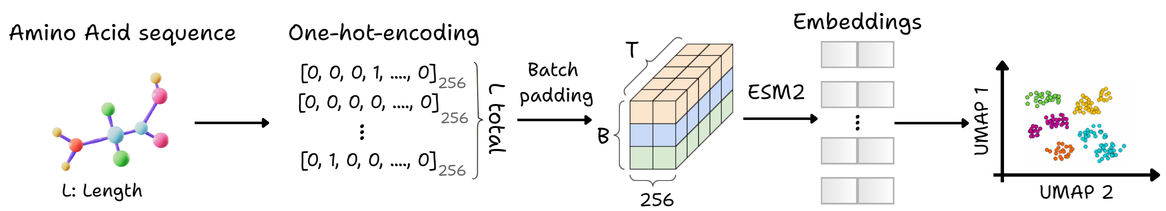
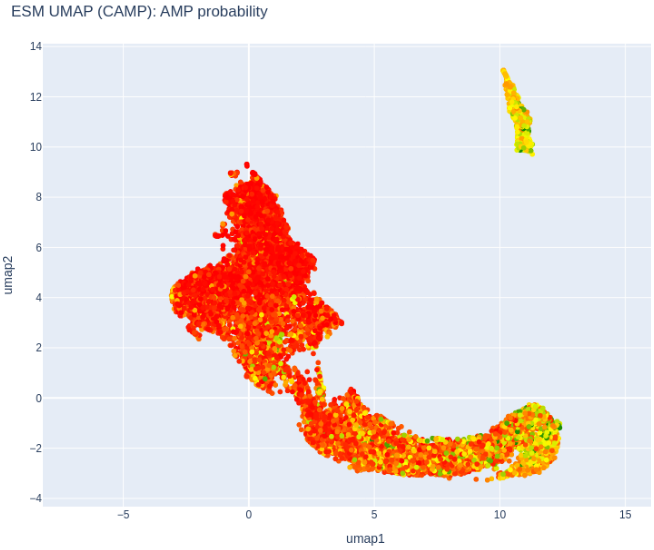
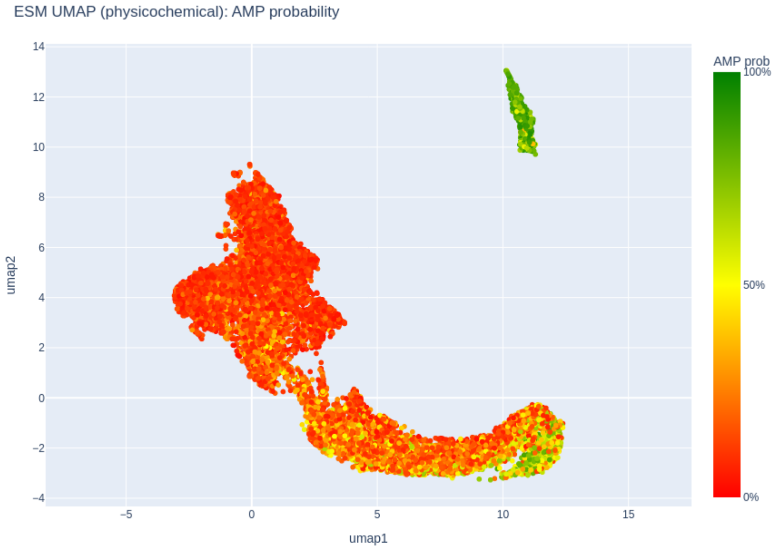


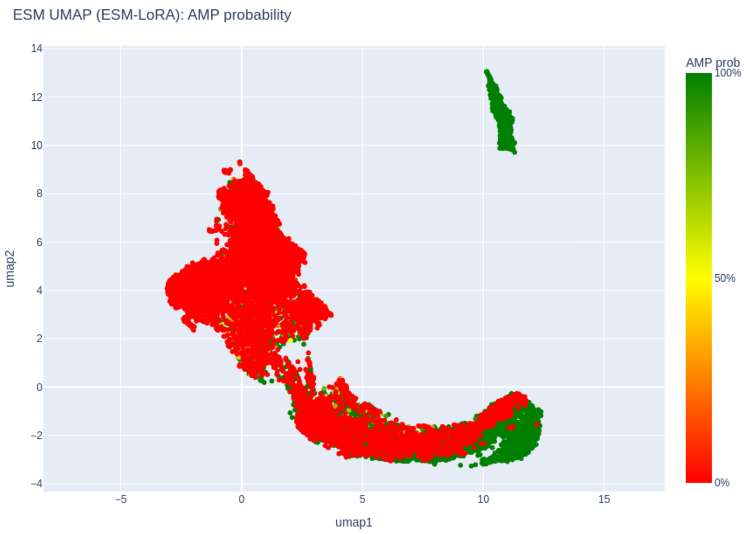
Figure 11: Pipeline for generating UMAP visualizations from amino acid sequences; where T is the length of the longest sequence in the batch and B is the number of sequences in the batch.



(a) CAMP-RF.



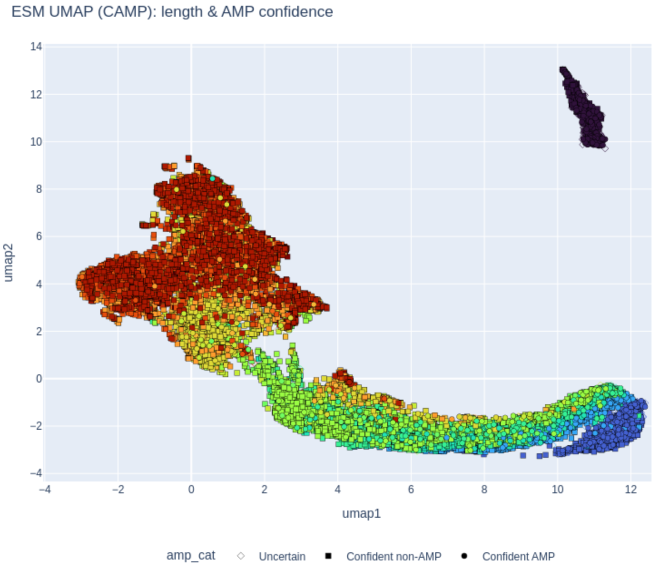
(b) Physicochemical.



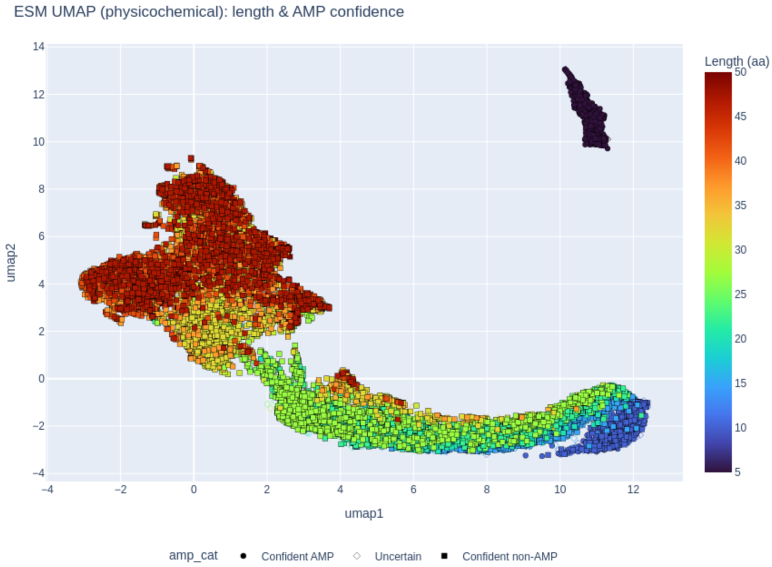
(c) LoRA–ESM.

Figure 12: UMAP projection colored by AMP–score for three evaluation methods.

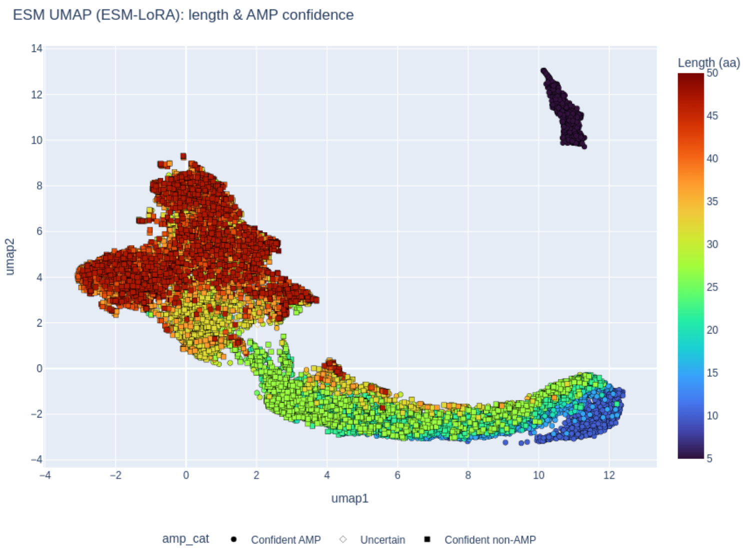
In the UMAPs above, colored by AMP score, we observe that the CAMP plot shows a scattered distribution with no clear separation between AMPs and non-AMPs. The physicochemical displays a similar layout, as we have on the bar-plots (Figs. 9, 8). In contrast, the LoRA–ESM plot is sharply polarized, with most points classified as either strong AMPs or non-AMPs, reflecting a binary decision boundary.



(a) CAMP-RF.



(b) Physicochemical.



(c) LoRA–ESM.

Figure 13: UMAP projection coloured by sequence length and AMP-confidence shapes for three evaluation methods; where ● = confident AMP, ⧫ = uncertain, ▼ = confident non-AMP.

Moving on, in these UMAPs, colored by sequence length, all three methods display the same color gradient, as they are based on the same embeddings. We can also observe how AMP classifications are distributed across different lengths, with confident AMPs and non-AMPs appearing at various positions depending on the evaluation method.

5 Ancestral Sequence Reconstruction (ASR)

In this study, we explore Ancestral Sequence Reconstruction (ASR) as a means to enhance the biological plausibility of protein generation. ASR allows us to infer the most likely ancestral protein sequences based on extant homologs, providing evolutionary context and insight into conserved and variable regions.

The primary goal of incorporating ASR is to identify uncertain positions within ancestral proteins, typically represented by ambiguous amino acid probabilities, and use this information to guide conditional generation. By masking these uncertain residues, we can employ EvoDiff to generate new protein sequences that are both evolutionarily grounded and structurally viable.

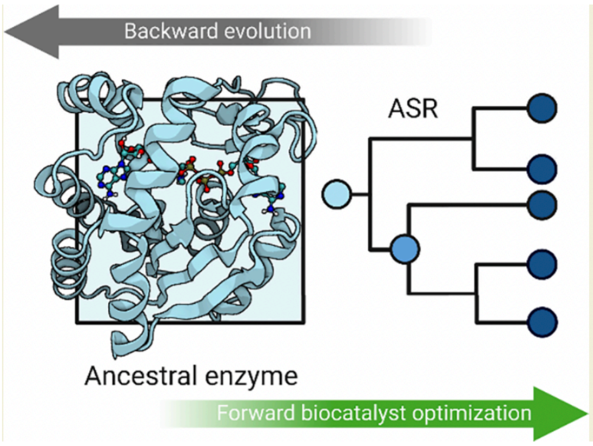


Figure 14: ***Visualization of ASR***: The process of tracing evolutionary history backward to infer ancestral proteins [61].

5.1 FireProt: Evolutionary Models using Maximum Likelihood Estimation

FIREPROT is a computational framework designed to engineer thermostable proteins by combining evolutionary insights with structural physics [55]. Starting from an MSA and a known 3D structure, FIREPROT employs maximum likelihood (ML) estimation to infer evolutionary constraints and identify stabilizing mutations.

Mathematically, FIREPROT estimates the change in folding stability, $\Delta\Delta G$, for each candidate mutation using energy-based predictors (e.g., FoldX, Rosetta) and evolutionary "back-to-consensus" scores derived from the MSA. Specifically, given a position i with observed amino acid frequencies $f_i(a)$, its consensus score can be expressed as:

$$S_{\text{evol}}(i, a) \;=\; \log f_i(a) - \log f_i(a_{\text{wt}})$$

where a_{wt} is the wild-type residue. FireProt then combines this score with predicted energy changes:

$$\Delta\Delta G_{\text{total}}(i, a) \;=\; w_{\text{evol}}\, S_{\text{evol}}(i, a) \;+\; w_{\text{energy}}\, \Delta\Delta G_{\text{energy}}(i, a)$$

Mutations with favorable $\Delta\Delta G_{\text{total}} < 0$ are considered stabilizing. FireProt further applies ML-based filtering, based on statistical models of evolutionary substitution.

Finally, multiple predicted stabilizing mutations are recombined under the assumption of additive effects. The resulting multi-point variants are evaluated structurally and energetically before experimental validation, enabling efficient design of thermostable proteins[6].

5.2 Phylogenetic Tree Inference

In this work, we investigate two protein sequences that are used as input for ASR.

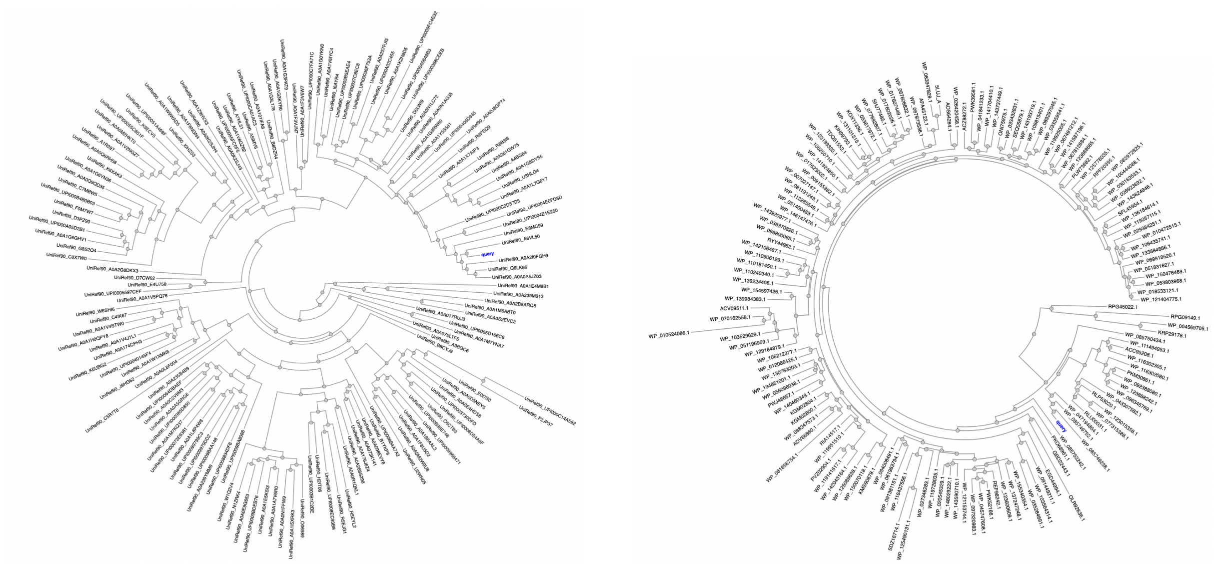
- **Escherichia coli maltose/maltodextrin–binding protein (MalE)** (UniProt ID: `P0AEX9`) is a periplasmic high-affinity receptor that captures maltose and higher maltodextrins and hands them to the *MalEFGK* ABC importer in *E. coli* K-12. The 396-residue precursor loses a 26-residue signal peptide, giving a 370-residue mature form.
- **Poly(ethylene terephthalate) hydrolase (PETase)** from *Ideonella sakaiensis* (UniProt ID: `A0A0K8P6T7`, `PETH_PISS1`) is a secreted serine hydrolase that depolymerises PET plastic to mono-(2-hydroxyethyl) terephthalate (MHET), terephthalic acid (TPA) and ethylene glycol, enabling the bacterium to use PET as a carbon source. The canonical sequence contains 290 residues, of which the first 33 constitute the Sec-type signal peptide; the mature catalytic domain therefore comprises 257 residues. Because of its biotechnological promise, PETase is the enzyme we target for stability engineering with FireProt.

In the figures below, we examine the phylogenetic trees generated by FireProt and focus on seven ancestral nodes for each protein. The objective is to identify uncertain amino acid positions in these ancestors and mask them in order to guide EvoDiff in completing the sequences with plausible, evolution-consistent residues.

[6]Thermostability is the ability of a substance to resist irreversible change in its chemical or physical structure.

Table 3: Length and ID of the two benchmark proteins.

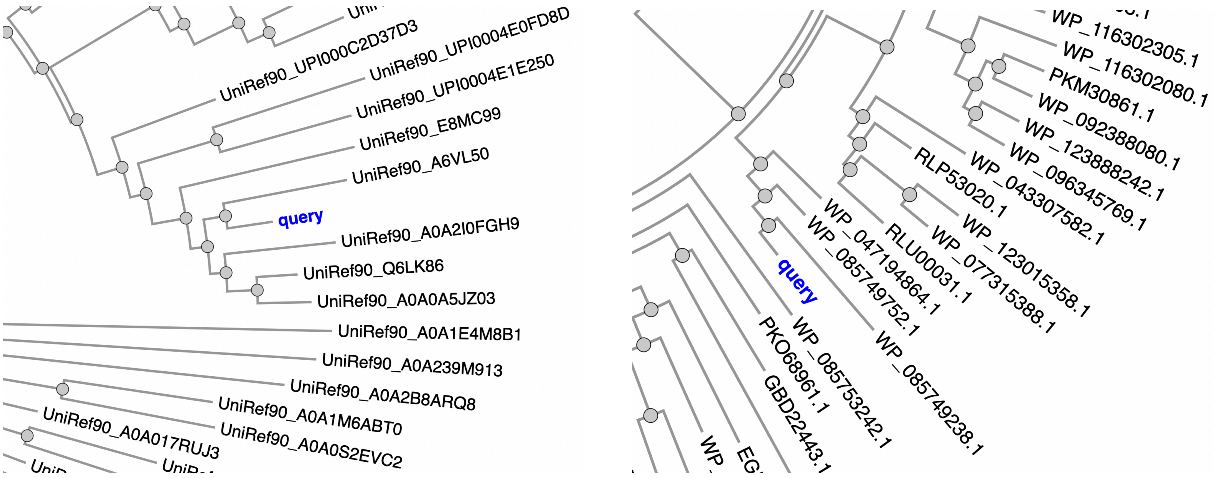
	MalE (*E. coli*)	**PETase (*I. sakaiensis*)**
UniProt ID	`P0AEX9`	`A0A0K8P6T7`
Precursor length	396 aa	290 aa
Mature length	370 aa (signal peptide 1–26)	257 aa (signal peptide 1–33)



(a) *E. coli*

(b) PETase (PETH-PISS1)

Figure 15: Maximum-likelihood phylogenetic trees.



(a) Zoomed-in *E. coli*. The query sequence is highlighted in blue.

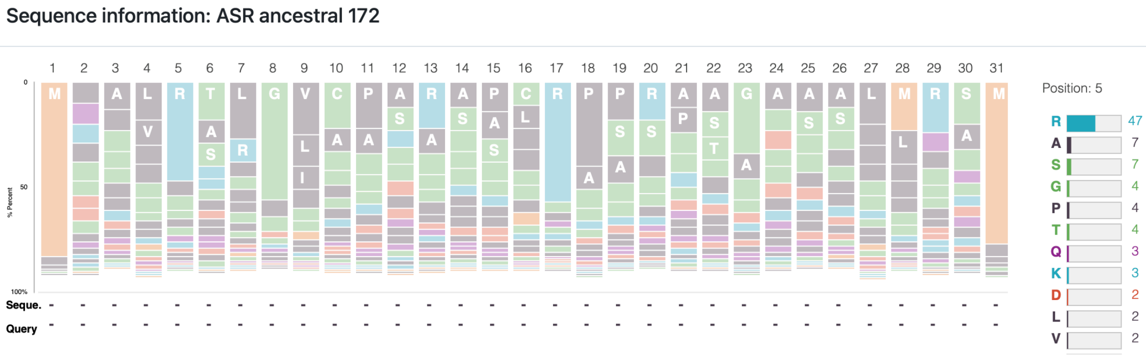
(b) Zoomed-in PETase (PETH-PISS1). The query sequence is highlighted in blue.

Figure 16: Maximum-likelihood phylogenetic trees (zoomed-in 🔍).

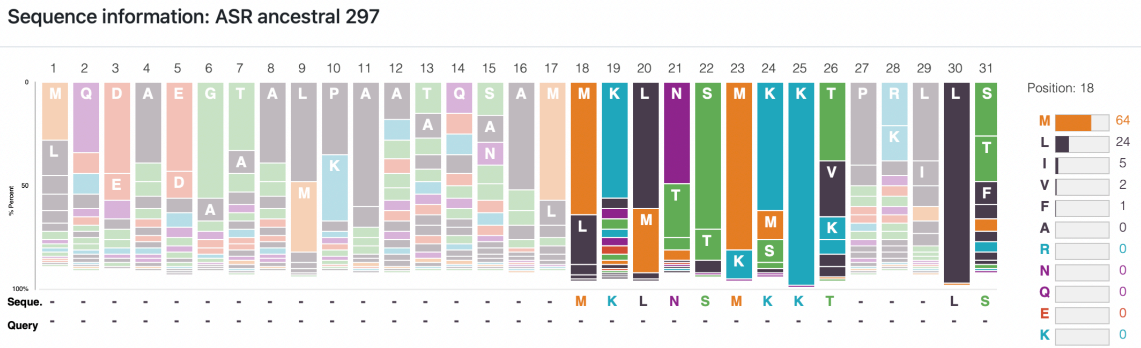
For each ancestral, FireProt computes site-specific posterior probabilities across the alignment.

5.2.1 Amino Acid Masking of Ancestors

The figure below shows the probabilities assigned by FIREPROT to each amino acid at every position for two specific ancestral sequences (in this case, the first ancestor of the root proteins we are studying).



(a) Posterior amino-acid probabilities for ancestor 297 (PETase family).



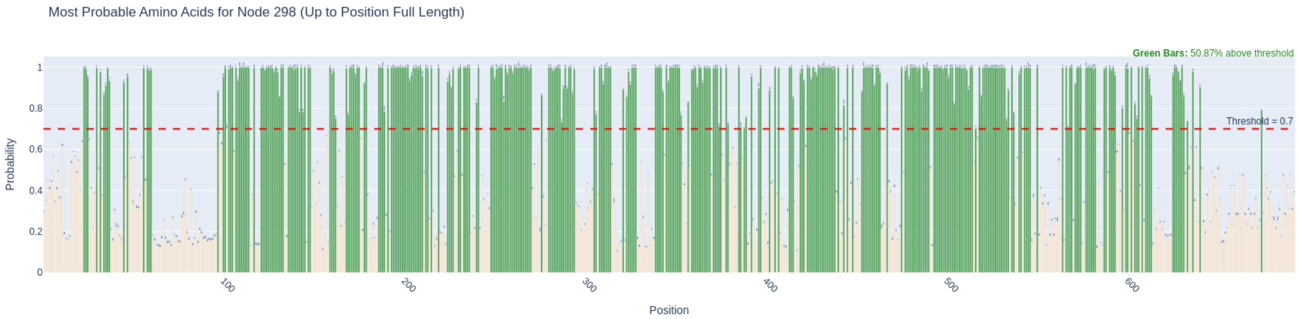
(b) Posterior amino-acid probabilities for ancestor 172 (MalE family).

Figure 17: ASR probability profiles. Bars represent the probability distribution across the 20 canonical amino acids at each position. Low-probability sites will be masked

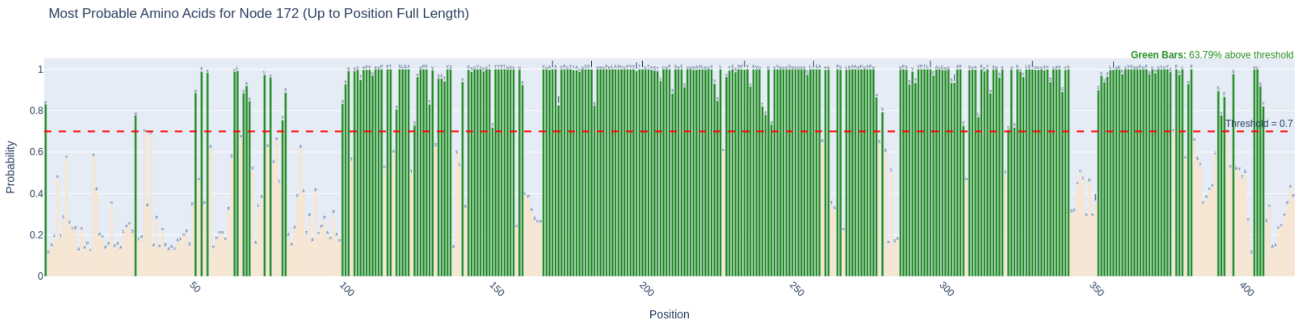
Positions are considered ambiguous, and thus masked, if the top amino acid probability is below ***70%***. Another criterion could be to mask positions where the difference between the highest and second-highest probabilities $\Delta p = p_{\text{top1}} - p_{\text{top2}}$ is less than 0.25. These masked residues are then provided to EvoDiff for conditional sequence generation, conditioning on the unmasked positions.

5.3 Conditional Generation using EvoDiff

For *E.,coli* MalE, we selected 7 ancestral nodes—297, 295, 292, 284, 280, 265, and 267—and conditionally generated 10 sequences per node by masking uncertain amino acid positions. Similarly, for PETase, we used 7 nodes—172, 171, 170, 157, 156, 155, and 152—applying the same masking strategy to produce 10 conditional sequences for each. The plots below show the percentage of masked amino acids per sequence and the pairwise similarities among the generated proteins.

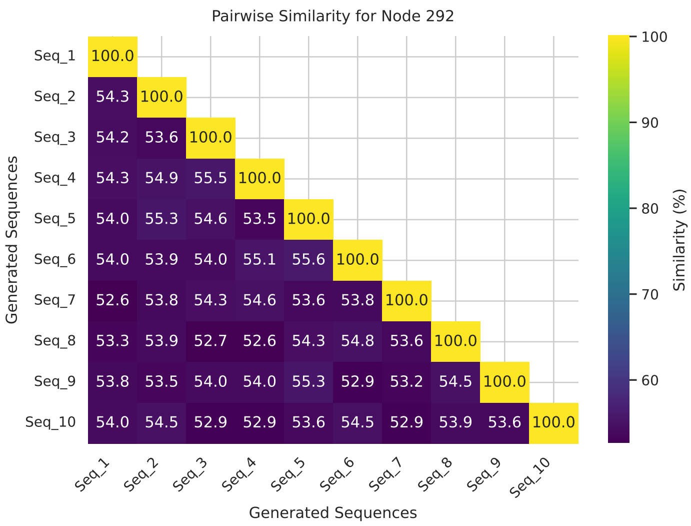


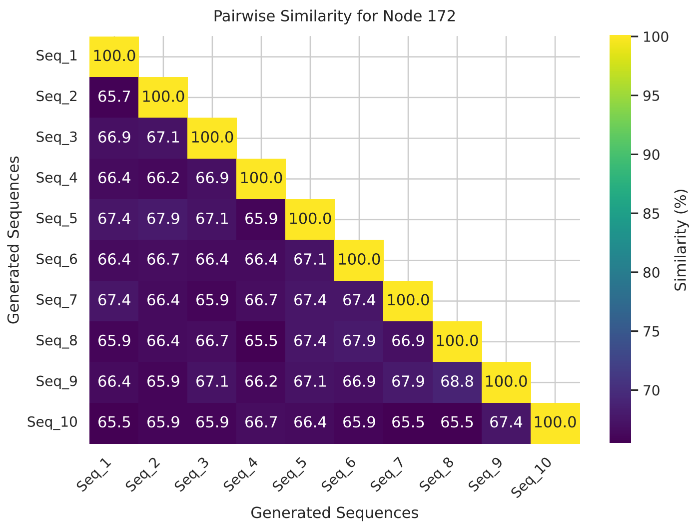
(a) MalE - Node 298; # masked amino acids ≈ 50%



(b) PETase - Node 172; # masked amino acids ≈ 36%

Figure 18: Most probable amino acids for each position; amino acids with probability <70% are masked.





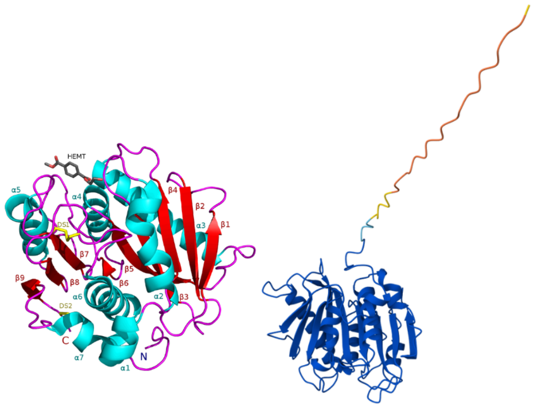
(a) MalE - Node 298; ≈ 55% on average, overall, and ≈ 9% on the masked amino acids.

(b) PETase - Node 172; ≈ 66% on average, overall, and ≈ 7% on the masked amino acids.

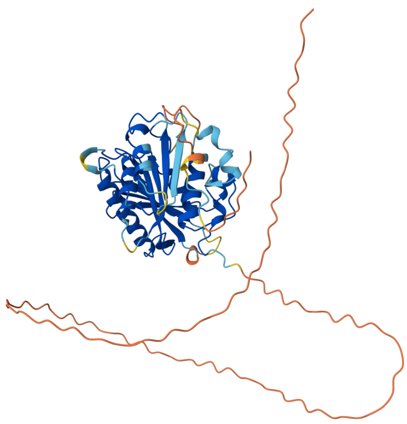
Figure 19: Pairwise similarities of generated proteins.

5.4 AlphaFold3 Structural Predictions

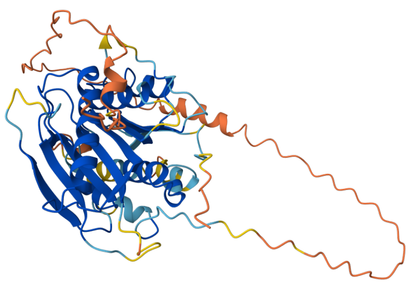
AlphaFold3 dramatically extends AlphaFold2 by predicting not only single-chain protein structures but also joint complexes with DNA, RNA, small-molecule ligands, ions, and post-translational modifications [9]. Its architecture replaces the Evoformer with a simpler "Pairformer" and employs a diffusion-based generative module to iteratively refine raw atomic coordinates, an approach borrowed from modern image-generation AI [9, 18]. This yields at least a 50% improvement in predicting molecular interactions overall and doubles accuracy in critical categories like protein–DNA and antibody binding [70, 9]. In the following figures, we utilize AlphaFold3 to predict the 3D structures of three generated proteins with high, medium and low sequence similarity to each root proteins.



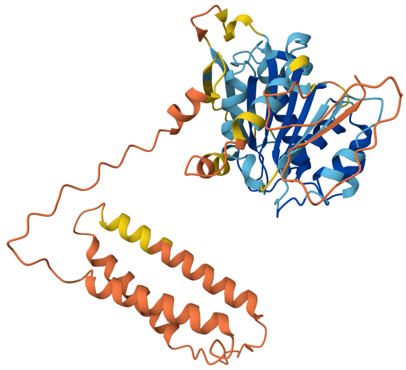
(a) PETase (PDB 5XH3, crystal structure); Ground truth (left) & AlphaFold3 (right)



(b) Most similar = 69.5%, pTM = 0.70 (Node 171).



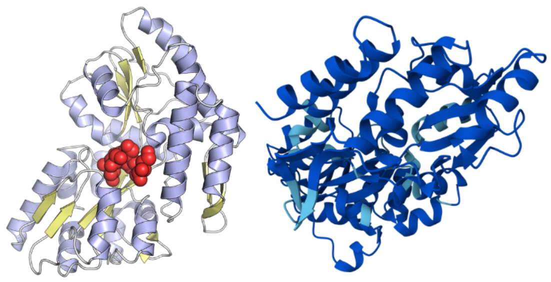
(c) Medium similarity = 64.5%, pTM = 0.71 (Node 157).



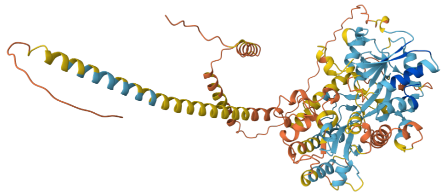
(d) Least similar = 42.7%, pTM = 0.61 (Node 152).

Figure 20: AlphaFold 3 predictions for PETase-related sequences compared with the experimental PETase structure.

plDDT: Very low (<50) Low (60) OK (70) Confident (80) Very hight (>90)

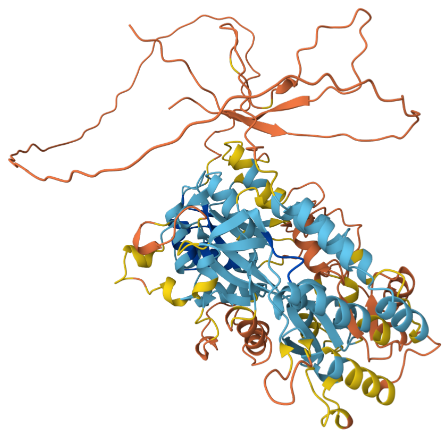


(a) E. coli MalE (PDB structure); Ground truth (left) & AlphaFold3 (right)



(b) Most similar = 57%, pTM = 0.58 (Node 280).

(c) Medium similarity = 54.5%, pTM = 0.56 (Node 297).



(d) Least similar = 50.6%, pTM = 0.60 (Node 267).

Figure 21: AlphaFold 3 predictions for MalE-related sequences compared with the experimental *E. coli* MalE structure.

Some structural observations. For **PETase panel (Figure 20)**, all three AlphaFold models recover the canonical PETase α/β-hydrolase core (parallel β-sheet flanked by helices, coloured cyan/blue for high pLDDT), even when overall sequence identity drops below 50%. The *most-similar* and *medium* variants (69.5% and 64.5% identity; pTM $\approx$ 0.70) overlay well with the crystal structure, while the *least-similar* protein (42.7% identity; pTM = 0.61) still preserves the catalytic core but acquires a long, low-confidence coil/helix extension (orange–red) that is absent from the experimental enzyme. Thus, higher sequence similarity mainly affects peripheral regions and disorder, whereas the active-site architecture remains intact across all three designs.

Moving on to **MalE panel (Figure 21)**, despite more modest identities (57% to 50%), AlphaFold reconstructs the characteristic two-lobe α/β fold of the *E. coli* maltose-binding protein. High-confidence regions (cyan/blue) correspond to the ligand-binding cleft observed in the crystal structure, whereas the largest deviations appear in flexible loops and elongated terminal helices coloured yellow–red. pTM values cluster near 0.58 ± 0.02, reflecting correct global topology with uncertainty in hinge orientation; even the *least-similar* model retains the core fold and hinge but shows the longest disordered tail. Overall, conditional generation preserves structural feasibility: core functional scaffolds are maintained, while sequence divergence is expressed mainly as low-confidence extensions rather than disruptive changes to the fold.

6 Conclusion

This study explored three complementary topics for expanding the protein design space with modern generative AI: latent-space interpolation, unconditional diffusion, and conditional diffusion guided by ancestral sequence reconstruction. Together, these approaches show how sequence generators and structure predictors can work in concert to propose functional yet diverse proteins.

Protein interpolation. By blending two distant proteins in ESM-2 latent space, we obtained a smooth structural trajectory from a β-sheet fold to an α-helical scaffold. Intermediate hybrids preserved catalytic motifs while revealing regions of flexibility and disorder, demonstrating that the interpolated sequence might contain the best of both worlds from two otherwise unrelated proteins. Notably, CAMP classified most of the interpolated proteins as AMPs with high probability.

Unconditional diffusion. Order-agnostic and discrete diffusion models produced thousands of AMPs. Multi-metric screening showed that short, cationic sequences are most frequently identified as antimicrobial, especially by a LoRA classifier, which likely over-weights length. CAMP and physicochemical scoring offered a more balanced view.

Conditional ASR generation. Starting from seven PETase and seven MalE ancestral nodes, we masked low-probable residues and in-painted them with EvoDiff. AlphaFold 3 predictions indicated that even the least similar variants retained the core folds and active sites, with divergence confined to low-confidence loops.

Generative AI has revolutionized protein engineering, replacing slow, trial-and-error methods with intelligent design, potentially solving major environmental challenges like plastic degradation, and enabling the combination of antimicrobial peptides with other functional proteins such as anticancer or antibacterial agents, with promising applications in ancestral sequence reconstruction, transforming the future of biotechnology.

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